

Explainable Artificial Intelligence and Pharmacogenomics for Opioid and Polypharmacy Risk Prediction: A Causal Modeling Approach using All-Payer Claims Data

2026-04-15

Project Summary / Abstract

This project addresses the *clinical decision gap* between high-performance predictive models and actionable point-of-care tools for opioid and polypharmacy risk. Utilizing the Virginia All-Payer Claims Database (APCD)—covering 6.9 million unique patients—we implement a partition-first architecture achieving a 15.1× throughput improvement. We apply a novel Consensus-Causal Filter (SHAP $\cap$ Formal Feature Attribution; FFA) to identify modifiable drivers of adverse drug events (ADEs). The project culminates in a privacy-first, serverless PGx Risk Dashboard providing real-time risk scores and counterfactual “What-If” analysis.

Project Narrative: This research moves beyond technically accurate models to provide user-centric explainability for vulnerable populations. By recovering pharmacogenomic (PGx) signals from administrative claims, we provide a scalable framework to reduce preventable emergency room visits associated with polypharmacy and the opioid epidemic.

Specific Aims

Overall Objective: Apply a standardized CRISP-DM pipeline and the Clinical OODA Loop (Observe–Orient–Decide–Act) framework to develop and evaluate a “causal calculator” for opioid and polypharmacy risk prediction from All-Payer Claims Data.

Long-Term Goal: Produce a replicable, causally grounded, and clinically actionable pipeline from raw claims data to bedside pharmacogenomic decision support, suitable for adoption by community pharmacies and Safety Net hospitals. The serverless architecture is designed to accept EHR-derived inputs as an optional future extension, though claims data alone are sufficient for full functionality.

Aim 1 (RQ1) — Identify Causal Drivers of Opioid-Related Emergency Department Visits. Pilot ensemble models (CatBoost/XGBoost/XGBoost-RF) have been trained across seven age bands (13–114), achieving AUROC ≥ 0.95 and PR-AUC ≥ 0.84 on 2016–2018 cross-validation sets. The Consensus-Causal Filter (SHAP $\cap$ FFA) has isolated modifiable causal features from correlated documentation artifacts. Pilot DTW clustering has already identified two trajectory archetypes — *Rapid-Onset* (4.2-month escalation window) and *Chronic-Escalation* (22.1-month window). The deliverable is prospective validation of all models and archetypes against the strict 2019 holdout to confirm generalizability and guide the timing of clinical interventions.

Preliminary Results: AUROC ≥ 0.95 and PR-AUC ≥ 0.84 across all seven age bands on internal cross-validation; DTW archetypes reproduced across three independent cohort splits. *Deliverable:* Prospective 2019 holdout validation and archetype concordance confirmation.

Aim 2 (RQ2) — Develop a “Causal Calculator” for Polypharmacy Interaction Risk. FFA multi-feature interaction testing has already identified 115 synergistic drug-drug interactions in geriatric populations, including Acetaminophen + Levofloxacin (IE = 16.3) and Levofloxacin + Lorazepam (IE = 11.9), invisible to standard pairwise DDI databases. Intervention Rate (IR) scores have ranked deprescribing priorities (simvastatin IR 7.0×10^{-4} , furosemide IR 2.0×10^{-4} , alprazolam IR 1.0×10^{-4}). Z-code monitoring analysis has identified a U-shaped effect distinguishing protectively monitored (Q2; OR = 0.25) from fragmented-care (Q4; OR ≈ 0.94) polypharmacy phenotypes. The deliverable is prospective holdout validation of IR rankings and peer-reviewed publication of the causal calculator framework.

Preliminary Results: 115 synergistic pairs and 5,021 high-risk triplets identified; IR-ranked deprescribing targets confirmed across three age bands. *Deliverable:* Prospective validation and dissemination.

Aim 3 — Translate Causal Rules to Privacy-First Clinical Decision Support. The serverless PGx Risk Dashboard has been deployed on AWS Lambda, bundling up to 84 per-bin ensemble models, a CPIC snapshot (573 Level A/B gene-drug pairs), and per-band density thresholds. *Imputation of Normality* maintains risk-band stability at $\geq 87\%$ of patients at 50% input sparsity. The stateless CPIC PGx card satisfies HIPAA minimum-necessary (45 CFR § 164.502(b)) architecturally.

Preliminary Results: Warm inference ≤ 100 ms; cold-start $\leq 3,000$ ms; 100% CPIC concordance across 573 test cases; risk-band stability $\geq 87\%$ at 50% input sparsity confirmed in load testing. *Deliverable:* Ongoing reliability monitoring, EHR input extension, and peer-reviewed publication of the deployment architecture.

Research Strategy

Significance

The Clinical Decision Gap. Current medication safety tools rely on simple counts and pairwise drug-drug interaction (DDI) checks, missing the nonlinear risk escalation of complex polypharmacy regimens. High-performance machine learning (ML) models for opioid and ADE risk have been published extensively, yet fewer than 10% of FDA-cleared AI/ML devices include prospective clinical validation evidence, and fewer still report deployment architecture sufficient for independent replication^{1,2}.

The Triple Gap. Our PRISMA-compliant Systematic Quantitative Literature Review of 5,839 candidate studies identified three co-occurring gaps in the existing literature: (1) the near-complete omission of pharmacogenomic (PGx) context from risk models; (2) the prevalence of temporal leakage—where future outcome information contaminates training features, producing logically inverted causal rules (e.g., buprenorphine predicting OUD)³; and (3) the failure to bridge validated models to deployment. This project directly addresses all three.

Population Need. Virginia’s opioid-related ED visit rate has increased by 34% since 2018. Geriatric polypharmacy (≥5 medications) affects 44% of adults aged 65+ and is the leading modifiable predictor of preventable hospitalizations. Standard pairwise DDI databases miss synergistic interactions involving three or more drugs—the very combinations FFA is designed to detect.

Innovation

This project introduces three conceptual and technical innovations:

- 1. The Consensus-Causal Filter (SHAP $\cap$ FFA).** A feature is designated causal only when independently confirmed by both distributional (SHAP) and structural (FFA decision-rule) analysis. This replaces heuristic feature selection with a two-method gate that demonstrably reduces false-positive causal claims.
- 2. Recovery of PGx Signal from Claims.** We demonstrate that CPIC Level A/B pharmacogenomic features are recoverable from administrative ICD/NDC codes at population scale, providing a scalable alternative to prospective genotyping for risk stratification in settings without access to genetic testing.
- 3. Privacy-First Serverless CDS.** The “compute, respond, discard” architecture satisfies HIPAA minimum-necessary without a Business Associate Agreement, enabling adoption by community pharmacies and Safety Net hospitals for whom institutional EHR integration is not available.

Approach

Data and Architecture

We utilize the Virginia APCD (1.8 TB; 2016–2019), covering 6.9 million unique patients annually, staged into a Bronze-to-Gold data lake via partitioned DuckDB workers. The partition-first architecture enforces strict temporal leakage prevention: a “B1” balancing constraint architecturally prevents any post-outcome features from appearing in training windows. The 2019 calendar year is held out prospectively for all model evaluation.

Fifteen Age Band $\times$ Cohort partitions (opioid-related ED: 13–64; non-opioid ED: 65–114) are processed in parallel, achieving 15.1 $\times$ throughput improvement over single-node processing. The PROBAST checklist was applied to ensure low risk of bias in predictor selection and analysis.

Model Selection for Tabular APCD Data

Although I have formal training in deep learning (BIOS 691), this project deliberately prioritizes tree-based gradient boosting (XGBoost, CatBoost) for high-dimensional tabular claims data. Comparative studies have shown that XGBoost outperforms or matches state-of-the-art deep architectures on most tabular benchmarks while requiring substantially less hyperparameter tuning⁴. A large-scale meta-analysis across 176 tabular datasets further found that gradient-boosted trees outperform neural networks on average, particularly on large and statistically irregular datasets with skewed or heavy-tailed feature distributions—properties characteristic of real-world claims data⁵. Given the additional requirement for stable, interpretable SHAP and FFA attributions, tree-based ensembles are

the most appropriate primary modeling choice. Models are tuned via Optuna Bayesian TPE and validated through Monte Carlo Cross-Validation (MCCV).

Aim 1 Implementation: Opioid Risk and Trajectory Archetypes

Ensemble risk scores (CatBoost + XGBoost + XGBoost-RF) are computed with performance-based weighting derived from MCCV PR-AUC and LogLoss on the 2016–2018 training set. The Consensus Filter selects features confirmed by both SHAP importance and FFA structural attribution. DTW clustering (Sakoe-Chiba band, $w = 20\%$) derives trajectory archetypes from 12-month pre-ED prescription sequences; archetypes are validated by 2019 holdout concordance and annotated with clinical intervention windows (30-day PDMP review for Rapid-Onset; physical therapy and mental health referral windows for Chronic-Escalation).

Mitigation of risks: If PR-AUC < 0.80 in low-prevalence bands, cost-sensitive weighting (class_weight proportional to prevalence) and SMOTE oversampling will be applied. DTW alignment will fall back to Euclidean distance if Sakoe-Chiba banding over-constrains short sequences.

Aim 2 Implementation: FFA Causal Calculator for Polypharmacy

FP-Growth (support ≥ 0.01 , confidence ≥ 0.50) generates pre-computed drug co-occurrence rulesets for the geriatric cohorts (65–74, 75–84, 85–114). FFA multi-feature interaction testing extends pairwise IE scoring to triplets, identifying synergistic combinations invisible to standard DDI databases. Intervention Rate (IR) scores rank deprescribing priorities by expected risk reduction per patient-exposure. Z-code monitoring status (ICD-10 Z79.x) stratifies patients by managed vs. unmanaged polypharmacy to identify fragmented-care phenotypes.

Mitigation of risks: If synergistic pair counts fall below 50, support thresholds will be lowered to 0.005. Z-code analysis is pre-registered as exploratory if cell counts prohibit propensity-matched comparison.

Aim 3 Implementation: Production Deployment and Validation

The AWS Lambda container bundles up to 84 per-bin models (3 algorithms $\times$ 4 density bins $\times$ 7 age bands), CPIC snapshot (573 Level A/B gene-drug pairs), and per-band density thresholds. Imputation of Normality fills missing features with training-set medians, maintaining risk-band stability at $\geq 87\%$ of patients at 50% sparsity. The CI/CD pipeline (GitHub Actions $\rightarrow$ ECR $\rightarrow$ Lambda) validates each deployment against automated PR-AUC ≥ 0.60 and LogLoss ≤ 0.50 gates.

Mitigation of risks: If Lambda container exceeds 10 GB, model compression (quantized XGBoost) and cohort-specific container sharding will be applied. PGx card generation falls back to diplotype-only output if PharmVar lookup fails.

Timeline

Year	Milestones
Y1	Aims 1 & 2 model training (<i>complete</i>); CH_1–CH_2 submitted (<i>complete</i>); Qualifying Exam (<i>in progress</i>)
Y2	Dashboard deployed (<i>complete</i>); CH_3–CH_5 submitted (<i>in progress</i>); dissertation defense (June 2026)

Research Training Plan

Overview

The training program is organized around four competency areas aligned with the Clinical OODA Loop (Observe–Orient–Decide–Act) framework for translational informatics. All coursework is completed; training activities now focus on independent research and dissemination.

Area 1: Advanced Claims-Based Pharmacoepidemiology (Observe)

This area builds foundational skills for managing the Virginia APCD and its 1.8 TB of longitudinal records.

Formal Doctoral Coursework:

- **HGEN 611 – Data Science I** (Fall 2018, B): Computational foundation for high-throughput genomic and claims-based data engineering.
- **HGEN 691 – Methods in Data Science** (Spring 2020, A): Advanced feature engineering and pipeline design for large-scale biomedical datasets.
- **BIOS 572 – Biomedical Data I** (Fall 2019, B): Statistical methods for large-scale biomedical datasets applied to ADE risk assessment.
- **FMBA 614/615/616 – Health Care Management I–III** (Summer 2017, A): Regulatory and systems-level context for healthcare delivery and outcomes research.

Professional Training: DAPT 611 (Analysis & Design of Database) and DAPT 614 (Advanced SQL) support the partition-first architecture; DAPT 621/622 (Statistics for Big Data) provide the quantitative foundation for APCD-scale feature analysis.

Area 2: Advanced Machine Learning and Causal Discovery (Orient)

This area focuses on moving beyond purely predictive “black-box” models toward causally interpretable ensembles supporting the Consensus-Causal Filter and IR scoring.

Modeling Mastery:

- **BIOS 691 – Deep Learning with R** (Summer 2020, A): Background in neural architectures and regularization. I use this to understand the broader model landscape and its limitations on clinical tabular data. Following evidence that tree-based gradient boosting (XGBoost, CatBoost) outperforms or matches deep learning on heterogeneous tabular datasets—especially large, skewed, and irregular datasets typical of health claims—I focus modeling on GBDTs integrated with SHAP and FFA rather than neural networks as primary models^{4,5}.
- **STAT 591 – Applied Bayesian Statistical Analysis** (Spring 2021, A): Theoretical basis for uncertainty quantification; motivates Bayesian TPE optimization (Optuna) for tuning tree-based ensembles.
- **DAPT 631 (Data Mining)** and **DAPT 632 (Forecasting Methods)** (Spring/Fall 2017, A): Applied predictive modeling for longitudinal clinical trajectories, informing DTW trajectory archotyping within a tree-based ensemble framework.

Applied Certification: AWS Certified Machine Learning – Specialty⁶ validates expertise in designing, deploying, and monitoring large-scale inference pipelines and stratified ensemble models across age bands and risk strata. Directly maps to Aims 1 & 2 (per-bin ensemble training and Optuna-tuned model selection) and Aim 3 (Lambda inference pipeline).

Area 3: Translational Informatics and Big Data Architecture (Decide & Act)

This area targets deployment of model insights as production-grade clinical decision support software.

Applied Certifications:

- **AWS Certified Solutions Architect – Associate⁷**: Provides the architectural foundation for deploying the serverless PGx Risk Dashboard on AWS Lambda with ephemeral, stateless compute, ECR container management, and CI/CD integration via GitHub Actions. Directly maps to Aim 3.
- **AWS Certified Big Data – Specialty⁸**: Validates expertise in orchestrating large-scale analytical workloads across S3 data lakes and partitioned DuckDB workers, enabling the 15.1× throughput improvement in the Bronze-to-Gold APCD pipeline. Directly maps to Aims 1 & 2 data architecture.

Genomic Foundations:

- **PHAR 691 – Advanced Pharmacogenomics** (Spring 2020, H): Biological context for recovering PGx signals from administrative claims.
- **HGEN 619 – Quantitative Genetics** (Fall 2019, C): Population-level genomic variation informing CPIC Level A/B feature engineering.

Applied Analytics: DAPT 612 (Text Mining), DAPT 613 (Business Intelligence), DAPT 615 (Emerging Technologies), DAPT 642 (Risk Analysis) provide the applied informatics breadth for full-stack clinical AI development.

Area 4: Geriatrics, Scientific Communication, and Research Ethics (Act)

This area supports transition to independent investigator and communication of XAI outputs to clinical stakeholders.

- **PHAR 691 – Geriatrics Research Group** (Fall 2023, P): Interdisciplinary clinical reviews validating causal calculator findings against geriatric prescribing standards (STOPP/Beers criteria).
- **Lean Six Sigma Black Belt** (Villanova): Managing reliability and safety of the clinical software lifecycle; reducing process variation in inference pipeline deployment.
- **DAPT 642 – Intro to Risk Analysis** (Spring 2017, A): Formal risk assessment integrated into the CI/CD pipeline validation gates.
- **FMBA 614–616 – Health Care Management**: Executive-level communication of complex XAI outputs to non-technical clinical leadership.

Fellowship Milestones

Completion of GRAD 691 (Dissertation Research) and all specialized PHAR/HGEN sequences positions me to finalize the Consensus-Causal Filter manuscript series and deploy the dashboard for the planned **June 2026 defense**.

Milestone	Target
Submit CH_1–CH_2 manuscripts	Q4 2025 (complete)
Doctoral Qualifying Examination	Q1 2026
Submit CH_3–CH_5 manuscripts	Q2 2026
Dashboard production deployment	Q1 2026 (complete)
Dissertation defense	June 2026

Candidate's Background and Goals

Personal Statement

My research goal is to bridge administrative claims data and clinical decision support using systems theory and military decision theory (OODA loops). As a Navy veteran and data scientist, I apply rigorous engineering discipline—enforced by Lean Six Sigma Black Belt training—to translational health informatics problems. The causal gap between high-performing models and bedside usability is the central challenge I address in this dissertation.

Preparedness

Quantitative foundation: Graduate training in Bayesian statistics (STAT 591, A), deep learning (BIOS 691, A), quantitative genetics (HGEN 619), and pharmacogenomics (PHAR 691) provides the methodological breadth for causal discovery in multi-omic, claims-based data. Prior M.S. coursework (DAPT series, all A) established mastery in database architecture, SQL, data mining, and forecasting—directly applied to the partition-first APCD pipeline.

Engineering credentials: AWS certifications (Solutions Architect, Machine Learning Specialty, Big Data Analytics) validate production-grade deployment competency. The PGx Risk Dashboard is currently live at sub-100 ms warm inference.

Demonstrated productivity: Five dissertation manuscripts are complete or in submission across four peer-reviewed journals (CTS, CPT:PSP, CPT), documenting a complete pipeline from APCD-scale data engineering to serverless CDS deployment.

Potential

The Clinical OODA Loop framework I developed integrates OODA decision theory with CRISP-DM methodology to create a reproducible, generalizable pipeline for translational clinical AI. This framework is designed to be replicable by any research team with access to an all-payer claims database and standard cloud infrastructure. The architecture is extensible to EHR-derived inputs as additional data sources become available, without requiring proprietary integration as a prerequisite.

Compliance and Data Management

Human Subjects

This project is a claims-based secondary analysis of de-identified records. **VCU IRB Protocol HM20022300** (non-human-subjects waiver) is in place, granted by the Virginia Center for Health Innovation (VCHI) Data Use Agreement. No identifiable patient data are collected, stored, or transmitted at any stage.

Privacy-by-Design

The PGx Risk Dashboard follows a “compute, respond, discard” pattern:

- Patient-linked inputs are processed in Lambda memory and discarded at function termination.
- No PII is written to S3, databases, or logs.
- The CPIC PGx card accepts only SNP variant data (no name, DOB, or identifiers); results are streamed directly to the client as a downloadable PDF.
- HIPAA minimum-necessary (45 CFR § 164.502(b)) is satisfied architecturally without a Business Associate Agreement.

Data Management and Sharing

Code, derived network feature specifications, and analysis scripts are shared via GitHub (github.com/Jerome3590/pgx-analysis) to ensure exact computational reproducibility. Manuscript submission packages (DOCX + TIFFs + supplementary tables) are maintained in the same repository. No individual-level claims data are shared; only aggregate model outputs and feature importance summaries.

Responsible Conduct of Research

Training focuses on: (1) data privacy and HIPAA compliance in AI healthcare applications; (2) ethics of algorithmic decision support for vulnerable populations; (3) reproducibility standards including pre-registered holdout splits and public code release; and (4) dual-use considerations for prescribing-behavior prediction systems.

References

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